



VOLLA COMMUNITY DAYS 2026

Oniro on Volla

An open mobile OS, running on **your hardware**

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WHAT IS ONIRO?

A mature, open OS platform — not a hobby project

OpenHarmony

Open-source OS launched in 2020 via Huawei's code donation to the OpenAtom Foundation — the open upstream of HarmonyOS.

Dual-foundation model

OpenAtom (China) + Eclipse (Europe) — global scale with regional trust.

Oniro

The Eclipse Foundation's downstream: the same technology, under European, vendor-neutral governance.

Already real

Production-grade frameworks, shipping at scale across phones, wearables, IoT and industrial devices.

1,800+

certified compatible products

13,000+

code contributors

140M+

lines of code

WHY IT'S DIFFERENT

One OS, from a sensor to a tablet

Multi-device by design

One codebase across Mini · Small · Standard — from MCU to wearable to phone & tablet

Distributed at the core

Devices discover each other and act as one "Super Device"

Modern app model

Declarative ArkUI — think SwiftUI / Compose / Flutter — with capability-based security

Purpose-built for devices

Not a desktop Linux retrofitted onto a phone

Mini — MCUs & sensors, ≥ 128 KiB

Small — cameras & routers, ≥ 1 MiB

Standard — phones & tablets, ≥ 128 MiB

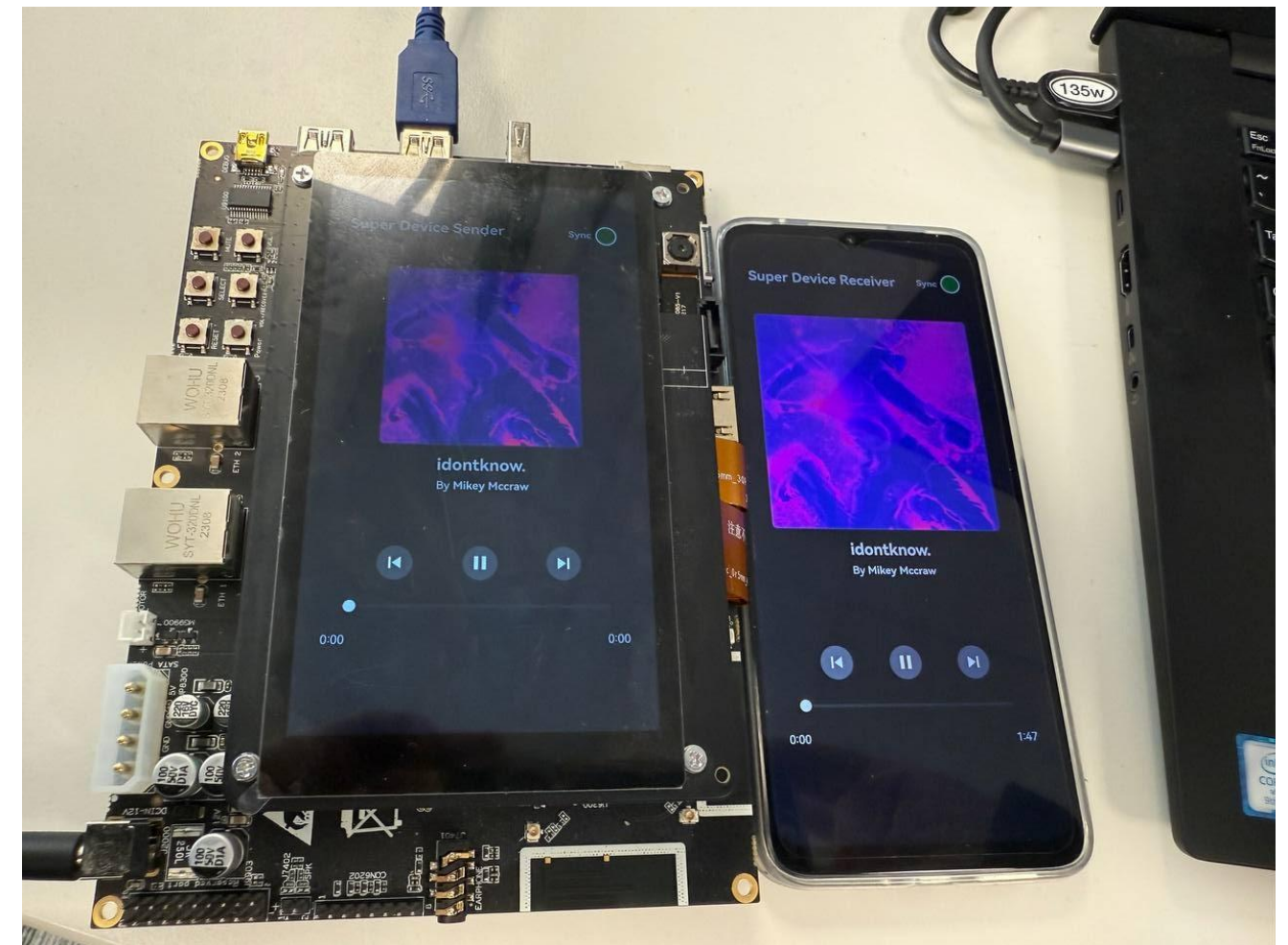
There are other open phones. Here's what's different.

	Ubuntu Touch	Droidian	Sailfish OS	Oniro / OpenHarmony
Base	Ubuntu	Debian	Mer/Nemo	Purpose-built OS
UI framework	Lomiri (QML)	Phosh (GTK)	Lipstick — partly closed	ArkUI (declarative)
App model	desktop-Linux + QML	desktop-Linux (apt)	QML + paid Android compat	HAP/Ability + RN · Capacitor · Tauri
Cross-device	—	—	—	DSoftBus / Super Device
Governance	UBports community	community	Jolla (commercial, closed UI)	Eclipse + OpenAtom — fully open
Backing scale	volunteer	volunteer	small company	Large open-source community (13,000+ contributors) + major-vendor R&D

Your devices, acting as one

- **DSoftBus** — zero-config discovery & secure connection between your devices
- **Data sync** — state and files follow you across devices
- **Remote app invocation** — start an *Ability* on another device
- **Device virtualization ("Super Device")** — borrow another device's camera, screen or speaker as if local

Continue a task phone → tablet · use the phone's camera as the tablet's · play audio on the nearest speaker



Super Device live: a dev board and a phone sharing one music session

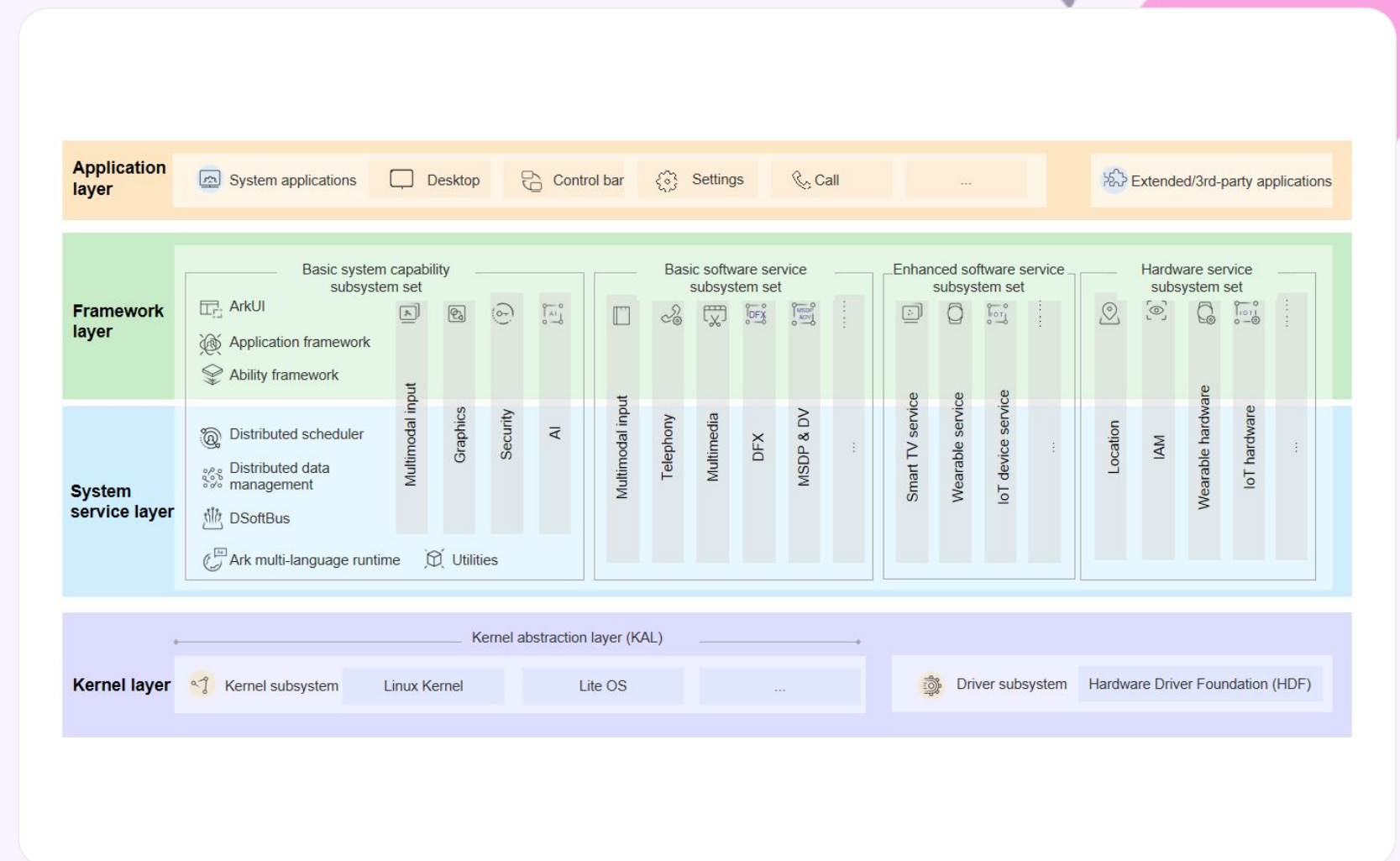
The kernel: it *is* Linux — plus a patchset

Multi-kernel: LiteOS (RTOS) for Mini / Small · **Linux** for Standard — phones & tablets

The Linux kernel = **LTS base (5.10 / 6.6) + OpenHarmony patchset** — *not vanilla mainline*

Required pieces: BINDER ASHMEM DRIVERS_HDF ACCESS_TOKENID SCHED_RTG

HDF (Hardware Driver Foundation) — a unified driver framework on top of the kernel, across LiteOS **and** Linux. It doesn't replace Linux drivers or the Device Tree — it abstracts driver logic & service publishing



OpenHarmony's layered architecture — one kernel abstraction layer spans Linux and LiteOS

Volla X23 reality: MediaTek MT6789 · LTS 5.10.209, rebuilt with the OHOS staging drivers (binder token-id, access_tokenid, hilog)

Three roads onto a real phone

A

Open BSP

the fully-open road

Mainline Linux + the OH patchset + **open drivers** : Mesa GPU, ALSA audio — the postmarketOS strategy. Done on the **OnePlus 6T** (Freedreno).

Fully open stack — where community mainlining has reached the SoC

B

OpenHarmony BSP

the manufacturer road

The SoC / device maker ships a BSP **built for OpenHarmony** : drivers & HALs linked against musl, plugged into HDF/HDI. How the 1,800+ certified products ship; reference boards like RK3568.

First-class, out of the box — needs the manufacturer on board

C

Android BSP + libhybris

the pragmatic road — Volla X23 today

Most retail phones ship **Android-only BSPs** — bionic HAL blobs; MediaTek mainline lags years behind. **libhybris** bridges musl ↔ bionic call by call, on the same Halium foundation as Ubuntu Touch / Droidian / Sailfish.

Unlocks thousands of existing phones today — blobs stay black boxes

Does OpenHarmony run on a mainline BSP? **Yes — road A, and it's the ideal where it exists.** Road B is what a manufacturer ships out of the box. Road C is how the X23 in your pocket runs it today — pragmatic, not ideological.

Native boot: OpenHarmony owns the device — PID 1

① The chainload

We reuse Halium's **signed boot.img** and swap only the ramdisk `/init`:
load vendor modules, mount the OHOS and Halium roots side by side,
exec OHOS init as PID 1.

② androidd

A ~300-line C launcher — deliberately *not* a port of LXC. It `clone()`s a PID
+ mount namespace, `pivot_root`s into the Halium root and starts only the
Android HAL services. **hwBinder** is **deliberately shared** between the two
worlds — that single shared binder device is the whole bridge.

OpenHarmony · PID 1

`init` → `samgr` → `render_service` → UI

Android HAL guest – androidd child

PID + mount namespace (Halium 12):

`hw servicemanager · composer@2.x · gralloc@4.0 (BT
/ sensors next)`

↕ **shared hwBinder**

Boot chain: MTK LK ▶ Halium `boot.img` (signed – kernel
untouched) ▶ `/init` = chainload (~150 LOC) ▶ OHOS `init`
· PID 1 ▶ `androidd` ▶ Android HAL services

A working stack — on devices never meant to run it

Natively, OHOS as PID 1 · Volla X23 (MT6789) + Volla Tablet (MT8781)



Graphics

Mali GPU via vendor blobs · render_service GPU composition

WORKING

Display & touch

720×1560 panel, brightness, multi-touch via HDF

WORKING

Audio

native ALSA — libasound + audio_host

WORKING

WiFi

HDI WPA path · association, DHCP

WORKING

Power

suspend / resume, charging, battery, power key

WORKING

USB / hdc

native ConfigFS gadget · live developer shell

WORKING

Cellular · camera · Bluetooth · sensors

modem / ISP / HAL integration — the active front

ONGOING

IN THE WILD

Putting this on Europe's open-source map

We're actively making open-source communities across Europe aware of this platform — and the opportunities it brings



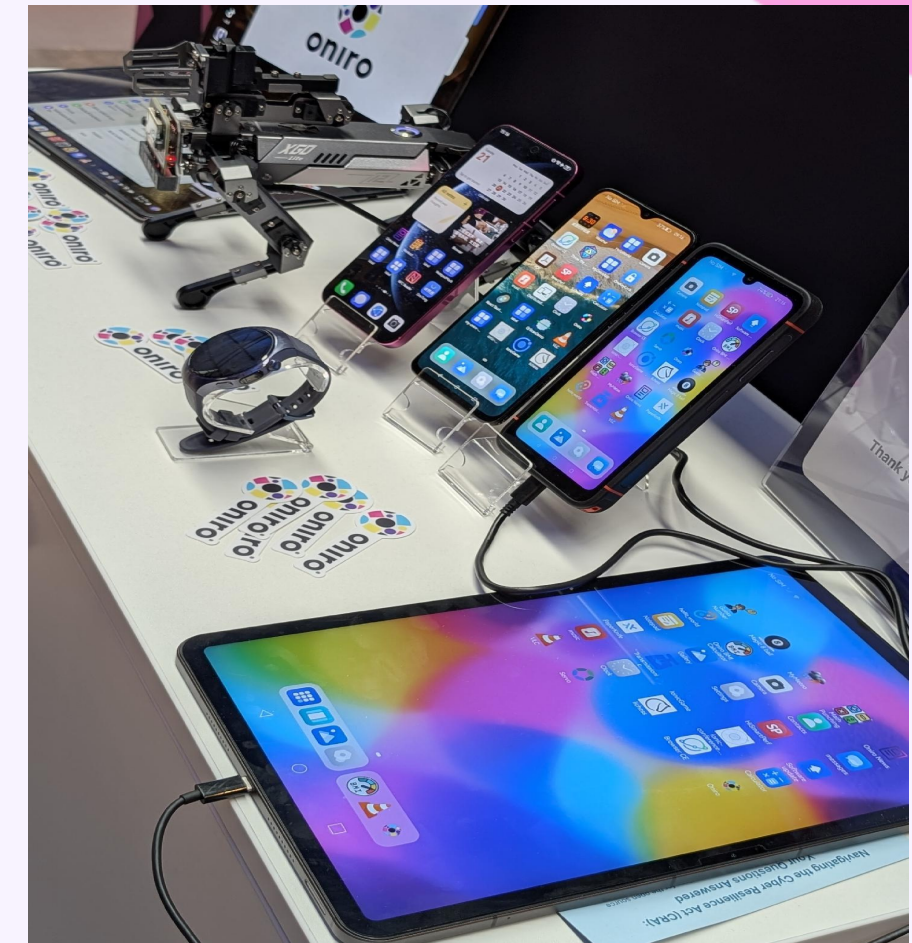
FOSDEM 2026 · Brussels

On stage — Bringing OpenHarmony to Phones



GOSIM Paris 2026

OpenHarmony Tech Forum — Lowering Every Barrier



OCX 2026 · Eclipse community

The Oniro booth — phones, tablet, watch, robot arm, all live

Real apps, three ways in — no rewrite required

A

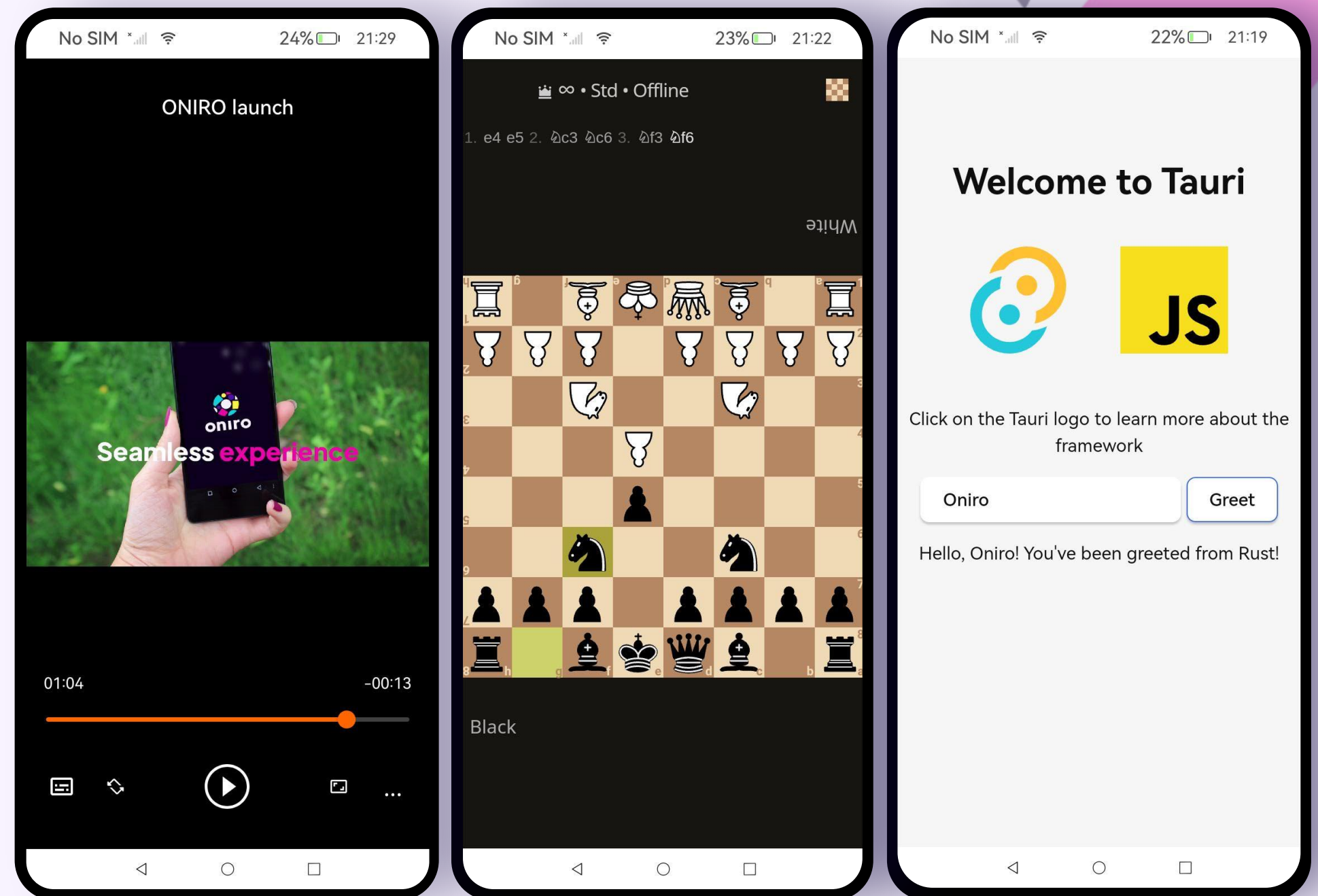
Native ArkTS

First-class apps. Flagship: **VLC** — the real libVLC core, 25 years of C/C++ and ~400 plugins, with a native ArkUI interface on top.

B

Cross-platform bridges

Capacitor (web) · **Tauri** (Rust) · **React Native** ; Flutter & Kotlin Multiplatform under research — thousands of existing apps, unmodified.



VLC (native ArkUI) · Lichess (Capacitor, unmodified) · Tauri starter (Rust) — running on Oniro today

Opening the app ecosystem to the world

Honest picture: the app ecosystem is large and real — but today it mainly covers the China region: catalogs, services, most of the developer base. Here's what we're doing about it.

LEVER 1 · BRIDGES

Near-zero port cost

The cross-platform bridges you just saw — Capacitor, Tauri, React Native — make bringing an existing app to the platform essentially free.

LEVER 2 · DEVELOPER ENABLEMENT

Zero to running app, in English

English-language docs (docs.oniroproject.org), templates and CLI tooling — English first.

LEVER 3 · DISTRIBUTION

No gatekeeper

Open distribution for this side of the world: a proof of concept today.

Where this goes next

01

Cellular & camera

From working bring-up to daily driver — modem and ISP integration

03

More phones & form factors

Including new Volla models

02

Bluetooth & sensors

Through the same androidd-resolved HAL path

04

Deeper libhybris coverage

Upstreamed, reusable by every port

GET INVOLVED

The doors are open — pick one

USERS · COMMUNITY

Try it on your Volla

Flash Oniro, break it, file issues, tell us what's missing — early-adopter feedback steers what we fix first.

DEVELOPERS

Bring a port — or an app

Your Haliu knowledge transfers directly to a device port. Or ship your app via Capacitor, Tauri or React Native — your codebase already runs.

PARTNERS · HARDWARE MAKERS

Bring Oniro to your hardware

Cooperate with us on adapting Oniro to your devices — a fully open OS platform, plus integration with a huge device ecosystem now starting to expand beyond China.

Come find me today — I have devices with me.



THANK YOU

Let's build an open, secure mobile ecosystem — together

Think **Global**, Code **Local**

CODE github.com/eclipse-oniro4openharmony

DOCS docs.oniroproject.org

PROJECT oniroproject.org

SPEAKER [Francesco Pham · LinkedIn](#)

